Loan Default Prediction

Comparing Classic ML and Neural Networks under Class Imbalance

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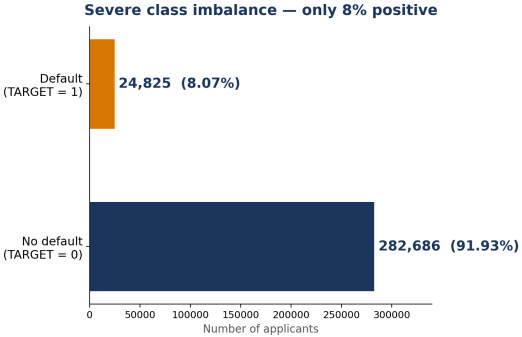
Final Project Group 15 · Home Credit Default Risk (Kaggle)

Problem & Motivation

Task	Binary classification — predict loan default ($\texttt{TARGET} \in \{0, 1\}$)
Data	307,511 applicants $\times$ 122 features (Kaggle)
Challenge	only **8.07%** default $\Rightarrow$ severe class imbalance
Approach	5 models (3 classic ML + 2 NN) + imbalance handling on best NN
Metric	ROC AUC (accuracy is misleading on imbalanced data)
Question	*Do classic ML and NN respond differently to imbalance handling?*

Dataset · Home Credit Default Risk

Structure	Tabular — row = one applicant
Observations	307,511 train / 48,744 test
Features	122 (102 numeric, 16 categorical)
Target	`TARGET` $\in \{0, 1\}$; 1 = default
Imbalance	**8.07%** default rate



Strongest signals

- `EXT_SOURCE_1/2/3` external credit scores
- Demographics, employment, credit bureau

Classic ML Models

Model	Best hyperparameters	Val AUC
Logistic Regression	`C`=0.1; `class_weight=balanced`	0.7487
Random Forest	`n_estimators`=200; `leaf`-regularized; `balanced`	0.7470
HistGradientBoosting	`lr`=0.05; `max_iter`=200; `leaf`=100	**0.7595**

Key takeaways

- HGBC wins by ∼1 AUC point — boosted trees dominate tabular.
- `class_weight='balanced'` on LR / RF handles imbalance in the loss.
- RF overfits at `leaf=1`: train 0.96 vs val 0.747.

Neural Network Models

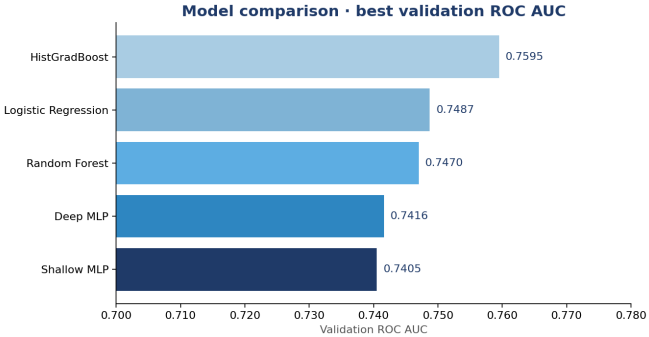
Architecture	Hidden	Parameters	Val AUC
Shallow MLP	$(64,)$	$\sim$15.7K	0.7405
Deep MLP	$(128, 64, 32)$	$\sim$42K	**0.7416**

ReLU · Adam · early stopping.
Tuned: `alpha` $\in \{10^{-4}, 10^{-3}\}$, `lr` $\in \{0.001, 0.005\}$.
Best for both: $\alpha = 10^{-3}$, `lr` $= 0.005$.

Key takeaways

- Depth doesn't help: 0.7405 vs 0.7416 (within noise).
- Tripling parameters bought nothing.
- Both NNs land below the classic learners.

Combined Leaderboard



Model	Val AUC
HistGradBoost	**0.7595**
Logistic Reg.	0.7487
Random Forest	0.7470
Deep MLP	0.7416
Shallow MLP	0.7405

Tree boosting wins by $\sim$1 AUC point.

Imbalance Handling on the Best MLP

Method	Thresh.	AUC	PR-AUC	Prec.	Recall	F1
Baseline	0.500	0.745	0.222	0.00	0.00	0.00
SMOTE	0.500	**0.648**	0.134	0.16	0.21	0.18
Threshold tuning $\star$	0.164	0.745	0.222	0.24	**0.40**	**0.30**

Key takeaways

- Baseline at 0.5: AUC 0.745 but recall **0%**.
- SMOTE hurts: AUC drops 0.10.
- Threshold tuning at 0.164 lifts recall to **40%**.

Threshold > resampling.

Conclusions

What we learned

- HGBC wins (val AUC = **0.7595**).
- NN ties LR / RF — depth alone doesn't help on tabular.
- Threshold tuning lifts MLP recall **0%** $\rightarrow$ **40%**; SMOTE hurt AUC.

Limitations

- Only main `application_*.csv` tables (auxiliary tables would add several AUC points).
- Single train/val split (PredefinedSplit), not full k-fold CV.

Recording

<<INSERT RECORDING LINK BEFORE SUBMISSION>>

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